

NEHA B

Email: nehareddy2401@gmail.com

Mobile: +12144585774

Data Engineer

PROFESSIONAL SUMMARY:

- Data Engineer with 4+ years of experience designing scalable cloud data solutions across financial and healthcare domains, emphasizing Microsoft Azure and distributed processing.
- Hands-on experience building scalable ETL and ELT pipelines using Azure Data Factory, Azure Databricks, and Azure Data Lake for reliable enterprise data delivery.
- Proficient in Python, PySpark, Spark, and SQL for large-scale transformations, data cleansing, profiling, reconciliation, partitioning, efficient joins, and performance optimization within Azure environments.
- Experienced developing parameterized Azure Data Factory pipelines and watermark-based incremental ingestion workflows, reducing unnecessary reprocessing while supporting scalable, reliable scheduled enterprise data refreshes.
- Strong experience implementing data quality frameworks using PySpark and SQL, including source-to-target reconciliation, schema validation, duplicate detection, business-rule validation, and controlled production reruns.
- Hands-on AWS data engineering experience using AWS Glue, Amazon S3, CloudWatch, Python, and PySpark for healthcare ingestion, transformation, monitoring, optimization, and analytical delivery.
- Experienced supporting Azure-focused and AWS cloud data platforms using Git, Agile practices, production monitoring, technical documentation, reusable ETL patterns, and analytics-ready dataset preparation.

TECHNICAL SKILLS:

- **Cloud Platforms** - Microsoft Azure, Amazon Web Services (AWS)
- **Azure Data Engineering** - Azure Data Factory (ADF), Azure Databricks, Azure Data Lake Storage (ADLS), ADF Pipelines, Linked Services, Datasets, Pipeline Parameters, Watermark-Based Incremental Loading
- **AWS Data Engineering** - AWS Glue, Amazon S3, Amazon CloudWatch
- **Big Data & Distributed Processing** - Apache Spark, PySpark, Spark SQL, Distributed Data Processing, Batch Processing, Large-Scale Data Transformation
- **Programming & Querying** - Python, PySpark, SQL
- **ETL/ELT & Data Integration** - ETL, ELT, Data Ingestion, Incremental Loading, Data Transformation, Source-to-Target Mapping, Data Cleansing, Data Profiling, Batch Data Pipelines
- **Data Quality & Validation** - Data Validation, Source-to-Target Reconciliation, Record-Level Checks, Schema Validation, Null Checks, Duplicate Detection, Business-Rule Validation
- **Performance Optimization** - Spark Optimization, Data Partitioning, Predicate Filtering, Efficient Joins, Incremental Processing, PySpark Transformation Optimization
- **Monitoring & Production Support** - ADF Pipeline Monitoring, Databricks Job Monitoring, Amazon CloudWatch, Logging, Root-Cause Analysis, Failure Recovery, Controlled Reruns
- **Development & Delivery** - Git, Version Control, Peer Reviews, Agile, Release Testing, Technical Documentation, Production Support

PROFESSIONAL EXPERIENCE:

Capital One

Jan 2024 – Present

Data Engineer

Responsibilities:

- Built scalable financial ingestion pipelines using Azure Data Factory, loading high-volume data into Azure Data Lake for downstream analytics and business reporting.
- Developed distributed transformations using PySpark and Azure Databricks, standardizing financial records and applying business rules to improve consistency across downstream analytical reporting datasets.
- Implemented incremental ingestion using Azure Data Factory parameters and watermark techniques, limiting unnecessary reprocessing and improving efficiency for recurring financial data refreshes daily.
- Optimized large-scale PySpark transformations using partitioning, filtering, and efficient joins, improving processing performance while preserving record completeness and required financial business logic.
- Created SQL reconciliation and record-level validation across source and target datasets, identifying discrepancies earlier and improving confidence in downstream financial reporting outputs.

- Profiled high-volume datasets using PySpark and SQL to detect nulls, duplicates, schema changes, and formatting issues before production processing and downstream consumption.
- Monitored Azure Data Factory pipelines and Databricks jobs through logs, diagnosed failures, and applied targeted fixes to improve reliability of scheduled data deliveries.
- Maintained pipeline code through Git, peer reviews, Agile releases, and technical documentation, improving change traceability and supporting consistent troubleshooting across production data workflows.

Labcorp

May 2022 – Aug 2023

Data Engineer

Responsibilities:

- Built automated healthcare data ingestion pipelines using AWS Glue, S3, and Python, improving data availability and reducing recurring manual processing effort for analysts.
- Developed PySpark transformations for structured healthcare datasets, standardizing fields and business rules to deliver consistent, analytics-ready data for reporting and operational business teams.
- Implemented automated data quality checks using Python and SQL, identifying incomplete records before downstream delivery and reducing recurring data reconciliation issues by 25%.
- Optimized AWS Glue processing through partitioning, predicate filtering, and efficient transformations, shortening scheduled pipeline runtime by 20% while preserving complete healthcare data outputs.
- Created incremental loading workflows using AWS Glue, S3, and change-based processing techniques, reducing unnecessary reprocessing and improving daily data refresh efficiency by 18%.
- Monitored production pipelines using CloudWatch logs and operational alerts, diagnosed failed jobs, and implemented targeted fixes that improved dependable dataset availability for analysts.
- Prepared curated healthcare datasets using PySpark and SQL for analytics and AI-related experimentation, improving data consistency, reproducibility, and usability without developing predictive models.
- Built reusable validation and reconciliation routines using Python and SQL, comparing source-to-target results and helping downstream teams identify data discrepancies before scheduled reporting.
- Collaborated with analysts and application teams using Agile delivery practices, translating healthcare data requirements into scoped pipeline changes that improved downstream dataset usability.
- Documented AWS pipeline dependencies, transformation rules, validation procedures, and recovery steps, reducing production troubleshooting time by 15% and improving engineering support handoffs overall.

Educational Details:

Master of Science - Advanced Data Analytics - University of North Texas

Bachelor of Technology (BTech) - CSIT - CVR College of Engineering